

A benchmark of cell-type annotation methods for single-cell data: cost, not accuracy, distinguishes them

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Abstract

Cell-type annotation by reference mapping is among the most frequently repeated operations in single-cell analysis, yet published comparisons report accuracy far more often than the wall-clock time and memory that decide what a working scientist can actually run. We benchmarked **thirteen methods** — classical, regularized-linear, parameter-free, correlation, deep-probabilistic, prototype-VAE and foundation-model — across **six datasets** (8–86 cell types; within-dataset, cross-dataset and cross-study splits) on commodity hardware, and validated externally on Open Problems **label_projection**, whose datasets, metrics and ranking we did not choose. Accuracy among the leading methods is tightly clustered — the top four span **0.008** — while their inference cost differs by two orders of magnitude and their peak memory by $2.5\times$. No method leads everywhere, and neither ordering is stable: accuracy inverts as the reference grows from 3k cells to a full atlas, and cost rankings invert with the input feature budget. Because several methods are now both accurate and cheap, the useful question is not which is best but what a low annotation cost makes possible. Using **actinn-jax**, a JAX reimplement of ACTINN with a cached train-once/map-many reference, we show that sub-second reference calling turns annotation from a single decision into a **multi-pass workflow**: a shipped ~ 800 -type reference routes a query to tissue and hands off to a focused reference (cross-study liver 0.23/0.58 to 0.72/0.86, exact/ontology); a pan-human annotator is distilled into an interchangeable entry point from raw counts alone, with no GPU and no labels, matching its teacher at six to nine times the throughput; and running three broad references over the same cells partitions them by agreement, where all three concur the concordance of every one of them roughly doubles. That partition costs three sub-second calls and no labels, which is what makes it usable on a query whose answers are unknown. We release the reimplement, the harness and the pre-trained references.

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1 Key Points

- Among leading annotation methods, accuracy differences are small (top four within 0.008) while predict time differs by $\sim 205\times$ and peak memory by $2.5\times$; cost, not accuracy, is what distinguishes them in practice.
- Rankings are not stable: a prototype VAE moves from worst to best as the reference grows from 3k to 49k cells, and a tuned linear pipeline that fits faster than a gene-space MLP on one panel costs $2.7\times$ more on another with a narrower feature budget.
- Predict is sub-second for every CPU method tested and flat in reference size for all of them; what makes chaining stages practical is the ratio, a sub-second call recurring against a fit of 19–123 s paid once.
- A pretrained pan-human annotator can be distilled into a fast reference using only raw counts

— no GPU, no labels — matching the teacher’s concordance and beating a census-built reference, at 6–9 $\times$ the teacher’s throughput.

- Agreement between independent broad references is a label-free confidence signal: on the 23% of cells where three references concur, each one’s concordance roughly doubles. A consensus *label*, by contrast, does not beat the best single reference.

2 Introduction

Annotating cells by mapping to a labeled reference is run constantly and rarely reported as a cost. Existing comparisons [Abdelaal 2019, Fu 2024] emphasize accuracy; the axis that decides what runs on a laptop — wall-clock and memory without a GPU — is usually absent. Foundation models [Kalfon 2025] raise accuracy in some settings but need accelerators, and their zero-shot label predictions underperform small models trained on curated references.

That reporting gap matters more than a missing column, because annotation in practice is not one decision. A working analysis asks what broad compartments are present, which tissue the query resembles, how the cells resolve against a reference matched to that tissue, whether anything is present that no reference describes, and how much of the answer to believe. Each of those is a reference call. When a call costs minutes, the workflow collapses to a single pass against a single reference chosen in advance; when it costs a fraction of a second, running several references and comparing them becomes ordinary. The benchmark below is therefore a means rather than an end: it establishes that the accuracy differences among the leading methods are small and the cost differences are not, which is what licenses spending the budget on more passes instead of on a better single pass.

Concurrent work sharpens the question rather than settling it. **Pan-human Azimuth** [Sarkar et al. 2026] ships a supervised hierarchical classifier over a harmonized organism-wide typology — 8 levels, 382 leaf types, $\sim$ 7M parameters over a fixed 5,055-gene panel, trained on 9.7M curated cells, with abstention *learned* rather than thresholded (expected calibration error 0.0044) — and runs on a laptop. It is better resourced than any reference we could build, and its authors reach a conclusion parallel to ours: training-data quality and organization matter as much as architecture or scale, with accuracy saturating past $\sim$ 5M training cells. A purpose-built pan-human model is therefore the right thing to *start* from; the open question is what to do next, since no fixed typology can re-annotate into a user’s own label set or resolve states below its own leaves.

Because a benchmark written by a method’s own author has a known failure mode, the comparison was constrained by construction: every method runs on **every** dataset through one harness on identical splits; the panel was chosen to include the baselines most likely to beat a small MLP, and each of them does beat it somewhere; our own classical tier is left untuned while the linear baseline is tuned; and the external validation uses a benchmark designed by others.

3 Materials and methods

actinn-jax. A dependency-light JAX reimplementation of ACTINN [Ma & Pellegrini 2020] — a 4-layer network (100/50/25 hidden units, ReLU, softmax, Adam) — replacing TensorFlow-1.x graph/session code that no longer installs on current Python and ML environments. Preprocessing is sparse-aware; a fitted reference is cached and reused across queries; prediction is chunked for atlas-scale inputs. Accuracy matches the original within repeat noise.

Panel and datasets. Thirteen methods (Supplementary Table S1) across six datasets (Supplementary Table S2): lung within-dataset and cross-dataset, liver within-dataset and cross-dataset, an 86-type blood+gut set, and PBMC — 8 to 86 cell types, spanning three generalization regimes.

Metrics. Accuracy, macro-F1, and **ontology-aware concordance**, which credits a call that is the same node, an ancestor or a descendant of the truth in the Cell Ontology. The last is required because vocabularies disagree about granularity: on our cross-dataset lung split reference and query share only 20 of 46 type names, so exact-match accuracy (~ 0.35 for every method) measures vocabulary mismatch rather than transfer. Concordance is reported only where both sides carry ontology ids.

Agreement between references. Three broad annotators answer in three different vocabularies, so agreement is defined in the Cell Ontology, where all three map: two calls agree when they are the same term or one is an ancestor of the other — the same relation the concordance metric uses. Cells are partitioned by how many of the three mutually agree.

Execution. Each method runs in its own environment as a separate process, because the dependency sets are mutually unsatisfiable; one driver builds each split once from a fixed seed and hands the identical pair to every method. Three repeats. Hardware: Apple Silicon, CPU for classical/linear/correlation tiers, Apple MPS for deep and foundation tiers. External validation runs on AWS `r7i.8xlarge` through Open Problems’ own Nextflow pipeline. Because wall-clock on a shared machine depends on co-scheduled load, cost is reported from the fastest of three runs on an otherwise idle machine, and external cost as a ratio to a method present in every run. Environments are pinned and the Cell Ontology release is recorded.

Supplementary material (separate document) carries the scaling studies (Figures S5–S7), the abstention sweep (Figures S8–S9), confusion matrices and per-class recall (Figures S1–S4), and Tables S1–S3.

Workflow components. A broad reference built from the CELLxGENE Census [CZI Census 2025]; a coarse→fine hierarchy obtained either from foundation-model embeddings or from Cell Ontology lineage; confidence-threshold abstention calibrated per reference by withholding 10% of cell types; masking-based refinement to a query’s own supported classes or to a tissue; and a cluster-level novelty screen. Protocols for each are documented in the benchmark repository.

4 Results

4.1 Accuracy is clustered; cost is not

The top of the accuracy table is a four-way cluster spanning **0.008**, led by a tuned linear pipeline rather than by a deep model (Table 1). Those same four differ by $\sim 205\times$ in **predict time** (0.33 s to 67 s) and $2.5\times$ in **peak memory**. Order within the cluster is not a result: the stochastic methods move by more than 0.008 between identical reruns, scANVI by up to 0.056, so the four are best read as tied on accuracy and separated by cost. `actinn-jax` holds the best ontology-aware concordance (0.811), likewise a margin inside repeat noise.

method	acc	macro-F1	ontology	fit (s)	predict (s)	peak mem (MB)
linear-anova-pca	0.839	0.699	0.808	3.4	0.33	4386
scArches	0.833	0.701	0.804	48.7	17.2	1773
scANVI	0.832	0.698	0.808	0.0*	66.7	2087
actinn-jax	0.831	0.683	0.811	24.2	0.54	2391
CellTypist	0.823	0.690	0.805	54.4	0.61	1569
SVM	0.808	0.675	0.796	55.2	0.07	1419
ProtoCloud	0.790	0.649	0.778	221.5	2.07	1907
SingleR	0.770	0.652	0.750	0.3	48.2	3612
kNN	0.770	0.623	0.768	0.4	0.19	1483
scTOP	0.739	0.619	0.703	1.1	1.21	1926
scmap-cluster	0.646	0.550	0.771	0.3	9.30	8609

Table 1. Accuracy and cost. Accuracy is the mean over the five shared-vocabulary datasets, macro-F1 over all six, and ontology concordance over the four that carry Cell Ontology ids; bold marks the best value in a column. *scANVI does most of its work in one train+predict pass, attributed to predict. Per-dataset scores: Supplementary Table S3.

Neither ordering is stable. Carrying four methods from a 3k-cell reference to a full atlas reverses the accuracy ranking — a prototype VAE moves from worst to best — and the external Open Problems panel, whose 1,000-gene budget is narrower than ours, reverses the cost ranking, with the tuned linear pipeline costing $2.7\times$ more than it does here (Supplementary Figure S5, and the extended report).

Prediction is where the annotation budget is actually spent, and it is small: sub-second for every CPU method tested, and flat in reference size for all of them, since a cached model’s inference does not depend on how many cells trained it. What makes chaining stages practical is therefore not a uniquely flat predict but the ratio — a sub-second call recurring against a fit of 19–123 s that is paid once (Supplementary Figure S6). With the reference held fixed, annotating a whole 525,000-cell atlas takes **41 s** (Supplementary Figure S7).

4.2 One query, two passes: routing and then resolution

That budget buys a workflow rather than a call (Figure 1). A shipped ~ 800 -type census reference annotates any query without being told what tissue it is; resolving those calls through the reference’s per-class tissue map identifies the tissue; a small focused reference for that tissue then re-annotates the same cells at full resolution. On withheld cross-study liver cells the broad pass scores **0.23 exact** / **0.58 ontology** and the focused reference reaches **0.72** / **0.86**.

The two passes hand off; they do not combine. Substituting a stronger broad model lifts the broad call but changes nothing downstream, and using it to narrow the focused pass’s classes makes the result **worse** ($0.731 \rightarrow 0.708$), because a wrong mask discards the correct class outright. Once the focused reference covers the tissue, the broad model’s value is routing, not resolution.

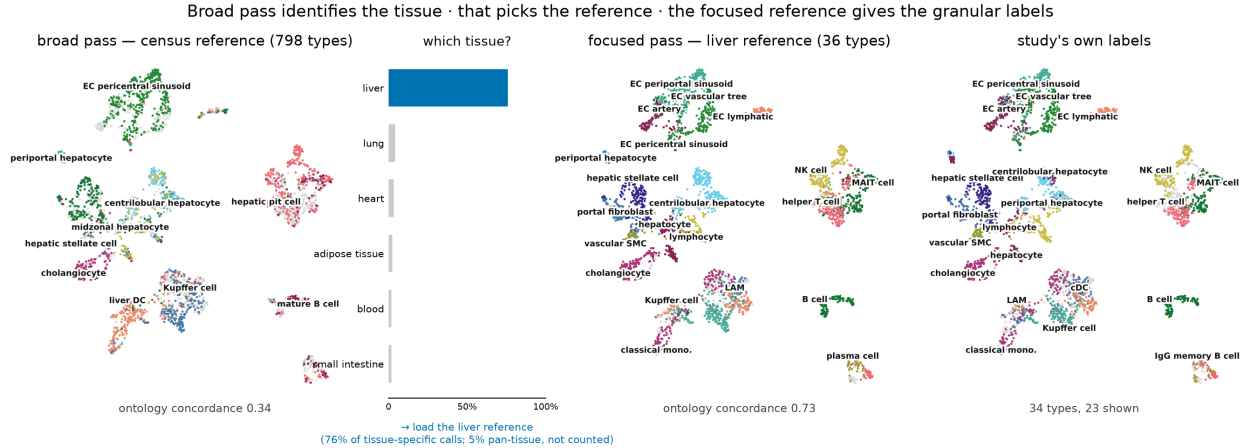


Figure 1. The workflow on a withheld liver study (3,396 cells), same embedding throughout. The census reference spreads 144 of its 798 labels over the query (concordance 0.34); resolving those calls to tissue gives **76% liver** against 4% for the next candidate, which selects the reference to load; the 36-type liver reference then re-annotates the same cells at **0.73**, tracking the clusters. Rightmost panel is the study’s own labels.

4.3 The broad entry point is interchangeable, and one can be distilled without a GPU

Building a broad reference from the census requires a foundation model on a GPU to discover its hierarchy. A pretrained pan-human annotator [Sarkar et al. 2026] already publishes one, so labeling a corpus with it and training on those labels transfers both vocabulary and structure — using **only raw counts**, no GPU and no labeled input, in under ten minutes of CPU. The result matches its teacher and beats our census-built reference, at six to nine times the teacher’s throughput (Table 2).

broad-pass model	classes	ontology	cells/s
census-built, ours	798	0.338	2,962
Pan-human Azimuth (teacher)	382	0.408	1,076–1,563
distilled from Azimuth, ours	324	0.406	8,937–10,021

Table 2. Broad-pass entry points on 3,396 withheld cross-study liver cells, all three scored on identical cells through one script. The distilled student needs only raw human counts to build and inherits the teacher’s vocabulary and hierarchy. It does not inherit the teacher’s calibrated abstention, which remains a limitation. Student and teacher are level here rather than separated: read the distillation as preserving the teacher’s annotations at a fraction of its cost, not as improving on them.

The GPU can be removed from the census route as well. Deriving the coarse hierarchy from Cell Ontology lineage rather than from a foundation-model embedding beats both no hierarchy at all (0.616 vs 0.547) and a same-sized random grouping (0.539), so it is the lineage structure and not the mere act of splitting that helps. Being species-independent, that route also reaches a second organism: a pan-mouse reference of 453 types across 85 tissues reaches 0.638 ontology concordance on two withheld datasets after 17 s of CPU.

4.4 Where independent references agree, all of them are right more often

Three interchangeable entry points invite a question that is only affordable when calls are cheap: what if a user runs all of them? Scored on identical cells and compared in the Cell Ontology, they partition the query by agreement, and the partition is informative (Figure 2). On the **23%** of cells where all three concur, every model’s concordance roughly doubles — the census reference from 0.338 to 0.690, the distilled reference from 0.406 to **0.778**, Azimuth from 0.408 to 0.785. On the 22% where none agree, all three fall to 0.21–0.28.

That all three improve together is what makes the signal useful: agreement is selecting cells that are unambiguous rather than cells that one model happens to get right. And unlike accuracy, it can be computed on a query whose answers are unknown, for the price of three sub-second calls and no labels.

A consensus *label*, by contrast, is not the prize. Taking the most specific call the agreeing references support scores **0.365**, below the best single reference’s 0.408, even after falling back to the strongest model where nothing agrees. Running several references does not produce a better annotation; it tells you which annotations to trust.

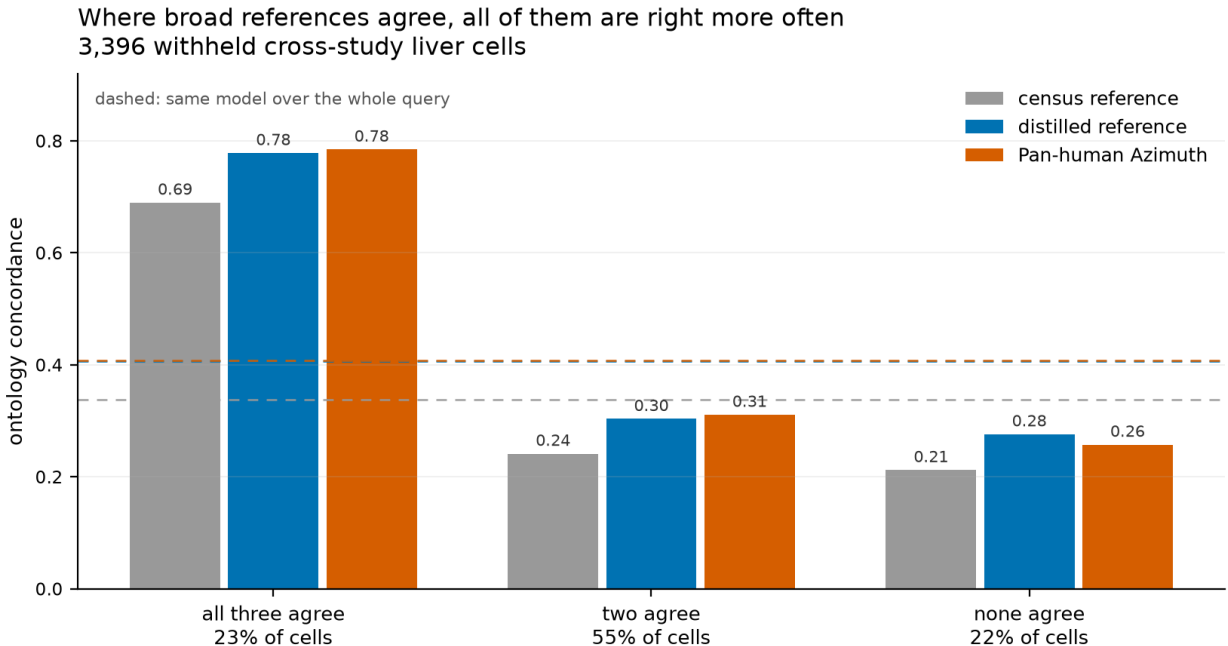


Figure 2. Ontology concordance within each agreement tier for three broad references over the same 3,396 withheld liver cells; dashed lines are the same models over the whole query. Agreement is evaluated in the Cell Ontology because the three answer in different vocabularies (798, 382 and 324 classes).

4.5 Abstention is tunable; zero-shot labels are not competitive

Withholding 9 of 36 cell types entirely and sweeping a confidence threshold over the eight methods that return a per-cell confidence, five trade coverage for accuracy across the range and three do not: CellTypist’s probabilities are saturated and give one operating point instead of a curve, scTOP’s projection score keeps only 6% of the query at $p \geq 0.5$, and ProtoCloud’s ambiguity flag is flat until

0.9 (Supplementary Figures S8–S9). Among the five that work, abstain quality does not separate them — actinn-jax and scArches are effectively tied (0.969 accuracy on 66% of cells with 73% of novel cells flagged, against 0.983 on 61% with 71%) — but cost does, at 0.54 s of predict time against 17.2 s. This is a second and complementary route to the same end as reference agreement: a calibrated threshold within one reference, or concurrence across several with no calibration at all.

Separately, a foundation model run zero-shot scored **0.201** ontology concordance against 0.917 for a reference-trained model on the same cells, taking $\sim 280\times$ longer — reproduced independently on Open Problems, where zero-shot entries sit at the bottom of the leaderboard.

5 Discussion

Across two independent benchmarks the same pattern holds: methods separated by less than a percentage point of accuracy are separated by two orders of magnitude in cost, and neither ordering is stable across reference size or feature budget. Reporting accuracy alone therefore under-determines the choice of method, and reporting it from a single reference size can invert the recommendation.

The consequence we think matters most is not a ranking but a change in what an annotation pipeline can afford to do. When a reference call costs a fraction of a second, the pipeline stops being one classification and becomes a sequence of cheap ones: route the query to a tissue, re-annotate against a reference matched to it, resolve states below the label, and ask several independent references where they agree. Each step is unremarkable alone; together they are what a user actually wants, and their feasibility is a cost property rather than an accuracy property.

The agreement result is the clearest case, and also the most honest about its limits. It buys no better label — the consensus call loses to the best single reference — but it identifies, with no ground truth at all, the fraction of a query on which independent references concur and on which every one of them is markedly more often right. That is available to anyone willing to run three annotators instead of one, which is only a reasonable thing to ask when each takes under a second.

We are explicit about what is not ours. ProtoCloud provides uncertainty, attribution and a retraining-based refinement path, and becomes the most accurate method at atlas scale. A purpose-built pan-human annotator is better resourced than our census-built reference, and our distilled model matches rather than improves on it; we claim not better annotations but cheaper ones that preserve its vocabulary and structure. What remains distinct is the hand-off no fixed typology performs — re-annotation into a user’s own label set, resolution below a typology’s leaves, and screening for what none of them claims — at a cost that keeps every stage on the machine already on the desk.

The main limitations are that the cross-method comparison is human-only and single hardware-family; that our classical tier is untuned while the linear baseline is tuned, which biases against our own method; that the distilled reference inherits a vocabulary but not its teacher’s calibrated abstention; and that the agreement result is one query in one tissue with three references, so its magnitude is not established beyond this setting. Full limitations are in the extended report.

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